

BENNETT JACKSON

SOFTWARE ENGINEER

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EDUCATION

Bachelor of Science in Software Engineering (May 2025) | University of Arizona | *Tucson, AZ* *Aug 2021 - Present*
GPA | 3.84

Awards | Dean's List with distinction (x3), Academic year distinction (x2), Engineering Honors Convocation Award (x2)

Extracurricular | Student advisory board for Software Engineering, Member of Software Engineering Wildcats Club

EXPERIENCE AND PROJECTS

Student Software Engineer | University of Arizona | *Tucson, AZ* *Jan 2024 - May 2024, Aug 2024 - Present*

- Developed a full-stack web application for the Systems and Industrial Engineering department to convert and visualize Bayesian networks from OML, enhancing researcher accessibility.
- Built a Flask backend with six RESTful API endpoints and containerized with Docker, ensuring seamless deployment.
- Created a responsive frontend with HTML, CSS, and JavaScript, boosting usability and satisfaction by 50%.
- Conducted thorough testing and debugging, reducing reported bugs by 95%.

Software Engineer Intern | Garmin | *Chandler, AZ* *May 2024 - Aug 2024*

- Developed safety-critical embedded C software for Garmin Flight Decks, conforming to DO-178 standards standards across 80+ aircraft models.
- Automated regression testing for 15 components using Python, enhancing the testing process.
- Identified and resolved seven bugs through software and requirement updates.
- Authored requirements for customer-requested features to ensure successful implementation and alignment with goals.
- Implemented six customer-requested features to enhance Garmin Flight Deck functionality.
- Leveraged software architecture to introduce runtime-determined features, boosting system flexibility and adaptability.
- Managed and tracked changes with Git, contributing 75+ commits to the master branch.

Software Engineer Intern | NASA Jet Propulsion Laboratory | *Pasadena, CA* *Jan 2023 - Jan 2024*

- Optimized Python scripts using various open-source libraries and APIs, reducing runtime by 32%.
- Led a team of three interns to develop an open-source software solution for application migration.
- Implemented CI/CD pipelines with Jenkins for automated testing and deployment of two pip packages.
- Managed and completed 90+ tickets and metrics across multiple sprints using Jira.
- Spearheaded internal customer-facing interactions to communicate project updates and address concerns.
- Utilized Git and GitHub for version control and collaboration, successfully merging 90+ pull requests to master.

Project | Weekly Playlist Manager

- Constructed in Python, this project leverages the Spotify API to automate playlist management, allowing users to retrieve and save songs seamlessly. Integrated CI/CD with GitHub Actions for regular updates and efficient deployment.

KEY SKILLS AND RELEVANT COURSEWORK

Coding Languages | C, C++, Python, Java, HTML, CSS, JavaScript, Verilog, Assembly, Bash, Swift

Technical Skills | Linux, Unix, Command-Line Interface, Git, GitHub, CI/CD, Jenkins, Docker, Maven, SQL, Spring, Spring Boot, RESTful APIs, JUnit, Unit Testing, Software Security, Data Engineering, ETL, Pandas, UML, OML, Flask, Scrum and Agile Methodologies, Object-Oriented Programming, Open-source Practices

Additional Skills | Creative Solutions, Strategic Thinking, Time Management, Software Development Lifecycle

Coursework | Software Architecture and Design, Software DevSecOps, Introduction to Cybersecurity Operations, Systems Cybersecurity Engineering, Software Assurance and Security, Software Requirements Analysis, Software Project Management, Introduction to Web Design and Development, Discrete Structures and Basic Algorithms, Computer Programming for Engineering Applications 1 and 2, Fundamentals of Computer Organization, Discrete Math, Digital Logic, Linear Algebra, Engineering Ethics